

SAP CPI Project Documentation

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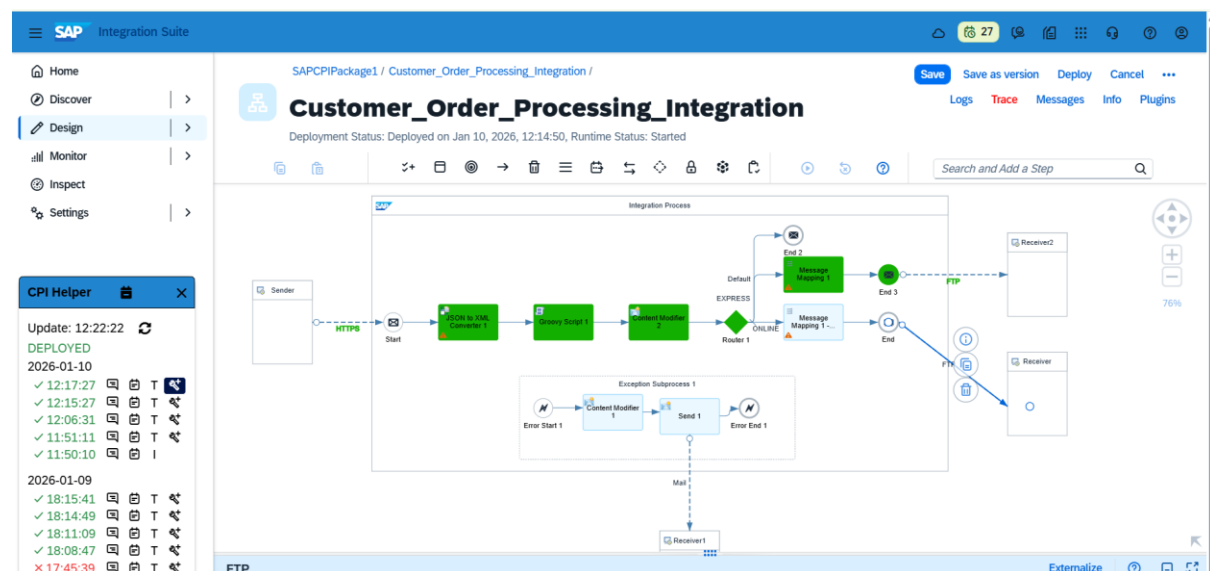
Date:29/12/2025

Scenario: Customer_Order_Processing_Integration

1.Scenario Overview

This project demonstrates a real-time order processing integration for an enterprise retail organization. Customer orders are received through a REST-based API in JSON format. These orders are processed in SAP Cloud Platform Integration (CPI), validated against business rules, and routed dynamically based on the order category (Online Store Orders or Express Store Orders). The integration ensures reliable processing with proper success responses and robust error-handling mechanisms.

2.Architechture Diagram



3. Step-by-Step Implementation

- Input Sales orders from customer received via a REST API (JSON payload).
- Implemented Groovy scripting to validate incoming request payloads, such as mandatory fields.
- Routing logic to differentiate and process Online Store Orders and Express Store Orders based on order type.
- Implemented success and failure response flows with meaningful HTTP status codes and response messages for API consumers.
- Exception Subprocess to manage error scenarios such as invalid payloads, missing fields, and routing failures.
- Gmail Notification will send to Support team in case of exception.

- Applied Logic, Business rules and posted to receiver system.

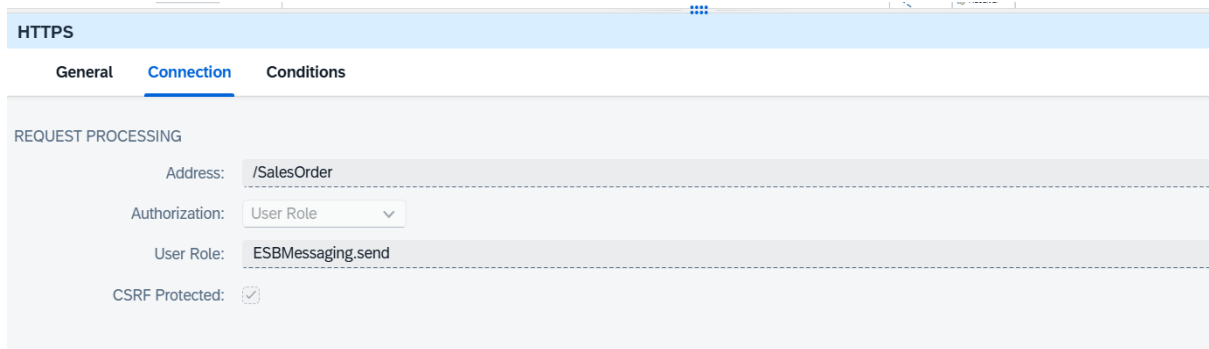
4.Adapter Configuration:

REST Adapter (Sender): HTTP Rest API

Endpoint: *https://b84a8864trial.it-cpitrail05-rt.cfapps.us10-001.hana.ondemand.com/http/SalesOrder*

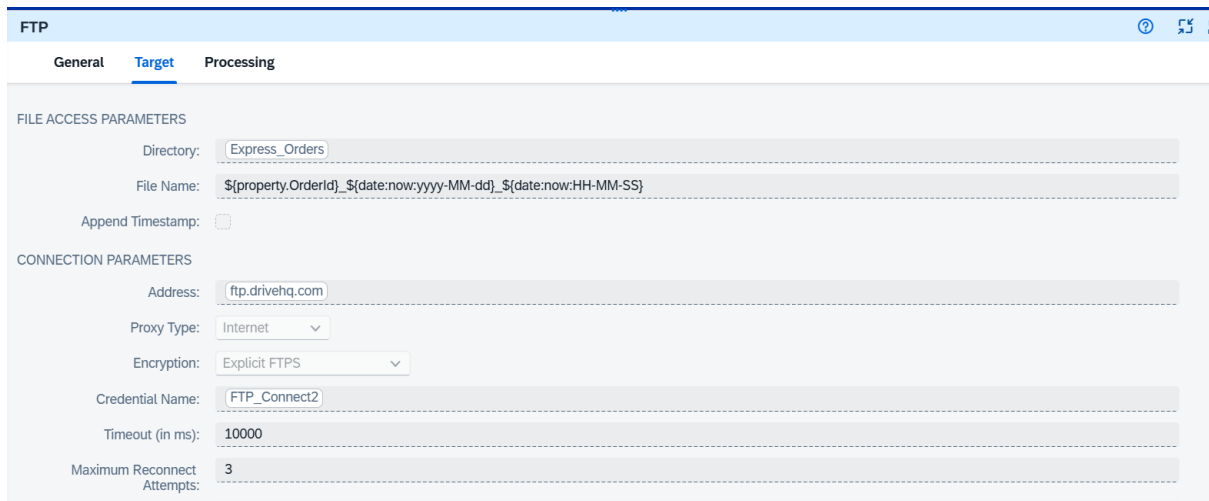
Method: POST

Authentication: Basic



The screenshot shows the configuration interface for an HTTPS adapter. The 'Connection' tab is selected. Under 'REQUEST PROCESSING', the 'Address' is set to '/SalesOrder'. The 'Authorization' is set to 'User Role' with a dropdown arrow. The 'User Role' is set to 'ESBMessaging.send'. The 'CSRF Protected' checkbox is checked.

FTP Adapter (Receiver):



The screenshot shows the configuration interface for an FTP adapter. The 'Target' tab is selected. Under 'FILE ACCESS PARAMETERS', the 'Directory' is set to 'Express_Orders'. The 'File Name' is set to a template: '\${property.OrderId}_\${date:now:yyyy-MM-dd}_\${date:now:HH-MM-SS}'. The 'Append Timestamp' checkbox is unchecked. Under 'CONNECTION PARAMETERS', the 'Address' is set to 'ftp.drivehq.com'. The 'Proxy Type' is set to 'Internet'. The 'Encryption' is set to 'Explicit FTPS'. The 'Credential Name' is set to 'FTP_Connect2'. The 'Timeout (in ms)' is set to '10000'. The 'Maximum Reconnect Attempts' is set to '3'.

5.Integration Components:

JSON to XML Converter:

JSON To XML Converter

General

Processing

Use Namespace Mapping: ☒

Namespace Mapping:

JSON Prefix	XML Namespace

JSON Prefix Separator:

Colon(:)

Add XML Root Element: ☒

Name:

SalesOrder

Namespace Mapping:

Content Modifier:

Content Modifier

General

Message Header

Exchange Property

Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	OrderId	XPath	//orderid	java.lang.String	
Create	OrderType	XPath	//ordertype	java.lang.String	

Router:

Router

General

Processing

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	ONLINE	\$(property.OrderType) = 'ONLINE'	☐
2	EXPRESS	\$(property.OrderType) = 'EXPRESS'	☐
3	Default		☒

6.Exception Subprocess: Handles errors and log details.

7. Groovy Script:



Integrations and APIs / SAP_CPI_Package1 / Customer_Order_Processing_Integration / script1.groovy /

script1.groovy

```
1 import com.sap.gateway.ip.core.customdev.util.Message
2
3 Message processData(Message message) {
4
5     // Read XML as plain String
6     String xml = message.getBody(String)
7
8     // ---- Mandatory Field Checks using String search ----
9     if (!xml.contains("<orderIds") || !xml.contains("</orderIds")) {
10         throw new Exception("Validation Error: OrderId is mandatory")
11     }
12
13     if (!xml.contains("<orderType") || !xml.contains("</orderType")) {
14         throw new Exception("Validation Error: OrderType is mandatory")
15     }
16
17     if (!xml.contains("<shipToParty>") || !xml.contains("</shipToParty>")) {
18         throw new Exception("Validation Error: shipToParty is mandatory")
19     }
20
21     return message
22 }
23
```

8. Input Data:

```
1 {
2     "orderHeader": {
3         "orderId": "SO-20260109-1001",
4         "orderType": "ONLINE",
5         "orderDate": "2026-01-09",
6         "salesOrg": "1000",
7         "distributionChannel": "10",
8         "division": "00",
9         "soldToParty": "CUST100001",
10        "currency": "INR",
11        "paymentMethod": "CARD",
12        "deliveryPriority": "STANDARD"
13    },
14    "orderItems": [
15        {
16            "itemNumber": "10",
17            "materialCode": "MAT10001",
18            "description": "Laptop",
19            "quantity": 1,
20            "uom": "EA",
21            "unitPrice": 65000
22        },
23        {
24            "itemNumber": "20",
25            "materialCode": "MAT10002",
26            "description": "Wireless Mouse",
27            "quantity": 2,
28            "uom": "EA",
29            "unitPrice": 1200
30        }
31    ],
32    "shippingDetails": {
33        "shipToParty": "CUST100001",
34        "deliveryAddress": {
35            "street": "MG Road",
36            "city": "Bangalore",
37            "state": "KA",
38            "postalCode": "560001",
39            "country": "IN"
40        }
41    }
42 }
43
```

9. Output Data:

```
<?xml version="1.0" encoding="UTF-8"?>
<SalesOrder>
  <OrderHeader>
    <OrderId>SO-20260109-1001</OrderId>
    <OrderType>ONLINE</OrderType>
    <OrderDate>2026-01-09</OrderDate>
    <SalesOrg>1000</SalesOrg>
    <DistributionChannel>10</DistributionChannel>
    <Division>00</Division>
    <SoldToParty>CUST100001</SoldToParty>
    <Currency>INR</Currency>
    <PaymentMethod>CARD</PaymentMethod>
    <DeliveryPriority>STANDARD</DeliveryPriority>
  </OrderHeader>
  <OrderItems>
    <Item>
      <ItemNumber>10</ItemNumber>
      <MaterialCode>MAT10001</MaterialCode>
      <Description>Laptop</Description>
      <Quantity>1</Quantity>
      <UOM>EA</UOM>
      <UnitPrice>65000</UnitPrice>
    </Item>
    <Item>
      <ItemNumber>20</ItemNumber>
      <MaterialCode>MAT10002</MaterialCode>
      <Description>Wireless Mouse</Description>
      <Quantity>2</Quantity>
      <UOM>EA</UOM>
      <UnitPrice>1200</UnitPrice>
    </Item>
  </OrderItems>
  <ShippingDetails>
    <ShipToParty>CUST100001</ShipToParty>
    <DeliveryAddress>
      <Street>MG Road</Street>
      <City>Bangalore</City>
      <State>KA</State>
      <PostalCode>560001</PostalCode>
      <Country>IN</Country>
    </DeliveryAddress>
  </ShippingDetails>
</SalesOrder>
```

10 Notification to Gmail in failure case:

The screenshot shows an email interface with a notification from 'rohitham2019@gmail.com'. The subject is 'SAP CPI Environment Error at Customer_Order_Processing_Integration'. The email body contains details about an integration failure, including an error message, correlation ID, message ID, and flow name.

SAP CPI Environment Error at Customer_Order_Processing_Integration Inbox x

rohitham2019@gmail.com Fri, Jan 9, 5:33 PM (20 hours ago) ☆ ☺ ↶ ⋮

to me ▾

Integration Failure Notification – Sales Order Processing

Dear Support Team,

The Sales Order integration has failed during inbound processing in SAP CPI.

Error detail: Cannot produce target element /SalesOrder. Queue has not enough values in context. Target xsd requires a value for this element, but target field mapping does not produce one. Probably the xml-instance is not valid to the source xsd, or the target field mapping does not fulfill the requirement of the target xsd.

Correlation ID: AGlg7oQW0oHcEXe3sOrciUGW2L_C

Message ID: AGlg7oTrcid8zyx6Dqq0tYjAyFR

Iflow Name: Customer_Order_Processing_Integration

Date: 09-01-2026 12:03 / SAP CPI - Tenant Number:

Thanks & Regards,

SAP CPI Team

11 Iflow Package:



Customer_Order_Processing_Integration.zip